

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Concept Mapping

Course Code:	CSA0311	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	Basics, Data, Data object, Data structure, Abstract Data Types (ADT), realization of ADT in 'C', Primitive and non-primitive data structure, Linear and non-linear data structure Arrays & Recursion, Performance analysis and Measurement: Time and space analysis of algorithms, Average, best and worst-case analysis.		
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Section / Batch:	AIML	Date:	

Code of Conduct

I, _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student _____

Instructions

- This test contains 2 concept mapping, Total: 100 Marks.
- Evaluation of Answer Quality -Checking whether the answer includes: Concept Identification , Linkages, Organization, Integration of Literature, and Clarity
- No DTP printouts or electronic devices permitted. They should provide materials in PDF format.

Rubrics

Questions	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Total Marks
Q1						
Q2						

Searching Algorithms

Searching Algorithm

Linear Search

- Sequential search
- checks each element

Time complexity

- Best case : $O(1)$
- Average case : $O(n)$
- worst case : $O(n)$

Suitable for small /
unsorted datasets

Efficiency

Better Performance

Binary Search

- Requires sorted Data
- Divides Data into halves

Time complexity

- Best case : $O(1)$
- Average case : $O(\log n)$
- worst case : $O(\log n)$

Suitable for large sorted
datasets

Smart city Traffic Management -

Concept map

